3.4.4 Fuels and human impacts on the atmosphere

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 7			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Crude oil is a mixture of a very large number of compounds. Crude oil is found in deposits underground, eg the oil fields under the North Sea.	Students will not be required to recall the name of any other fractions.	5.7.1.1	4.8.1.2
Crude oil may be separated into fractions by fractional distillation. This process, which takes place in a refinery, can be used to produce a range of useful fuels and oils, including petrol and diesel.		5.7.1.2	4.8.1.3
Outcome 8			
When fuels burn completely the gases released into the atmosphere include carbon dioxide, water (vapour), and oxides of nitrogen. Sulfur dioxide is also released if the fuel contains sulfur. When fuels burn in a limited supply of air carbon monoxide is produced. Solid particles (soot) may also be produced. Students may be required to describe the impact on the environment of burning fossil fuels. Oxides of nitrogen and sulfur dioxide cause acid rain and problems for human health. Carbon monoxide is a colourless, odourless, poisonous gas that can cause death. Solid particles can cause global dimming and problems for human health.	**Suggested activity for TDA** Compare the amount of soot produced when burning different fuels.	5.9.3.2	4.4.1.6
Outcome 9			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Some human activities increase the amounts of greenhouse gases in the atmosphere, such as carbon dioxide from burning fossil fuels and methane from landfill and cattle farming.		5.9.2.2	4.4.1.4
Increased levels of greenhouse gases in the atmosphere cause the temperature to increase. Many scientists believe that this will result in global climate change.		5.9.2.3	4.4.1.4